

## Lecture 9

# Regression (Cont.) and Bias-Variance Trade-off

More on probabilistic view of regression and bias-variance trade-off

**EECS 189/289, Fall 2025 @ UC Berkeley**

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# Roadmap

- MLE Recap: Least Squares as Maximum Likelihood
- Choosing Different Noise Models
- Prior Beliefs
- Bias-Variance Trade-off





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# MLE Recap: Least Squares as Maximum Likelihood

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# Least Squares $\hat{=}$ Maximum Likelihood

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

The optimum value of  $\mathbf{w}$  in the above equation equals the optimum value of the log of the function (because log is a monotonically increasing function)

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = \sum_{n=1}^N \ln(\mathcal{N}(t_n | \overset{\mu}{\mathbf{w}^T \phi(\mathbf{x}_n)}, \sigma^2))$$

Writing the equation for a Normal distribution

$$= \sum_{n=1}^N \ln \left( \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2} \right)$$

Applying  $\log a \times b = \log a + \log b$

$$= \sum_{n=1}^N \left( \ln \left( \frac{1}{\sqrt{2\pi\sigma^2}} \right) - \frac{1}{2\sigma^2} (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 \right)$$

Simplify and separate two terms

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2} E_D(\mathbf{w})$$



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# Least Squares $\hat{=}$ Maximum Likelihood

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2} E_D(\mathbf{w})$$

$$\nabla_{\mathbf{w}} \ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{1}{\sigma^2} \nabla_{\mathbf{w}} E_D(\mathbf{w}) \longrightarrow \mathbf{w}_{ML} = (\Phi(\mathbb{X})^T \Phi(\mathbb{X}))^{-1} \Phi(\mathbb{X})^T \mathbf{t}$$

$$\Phi(\mathbb{X}) = \begin{bmatrix} \phi(\mathbf{x}_1)^T \\ \phi(\mathbf{x}_2)^T \\ \vdots \\ \phi(\mathbf{x}_N)^T \end{bmatrix}$$

The least-squares solution is the **MLE** under Gaussian noise.

# Choosing Different Noise Models

- MLE Recap: Least Squares as Maximum Likelihood
- Choosing Different Noise Models
- Prior Beliefs
- Bias-Variance Trade-off





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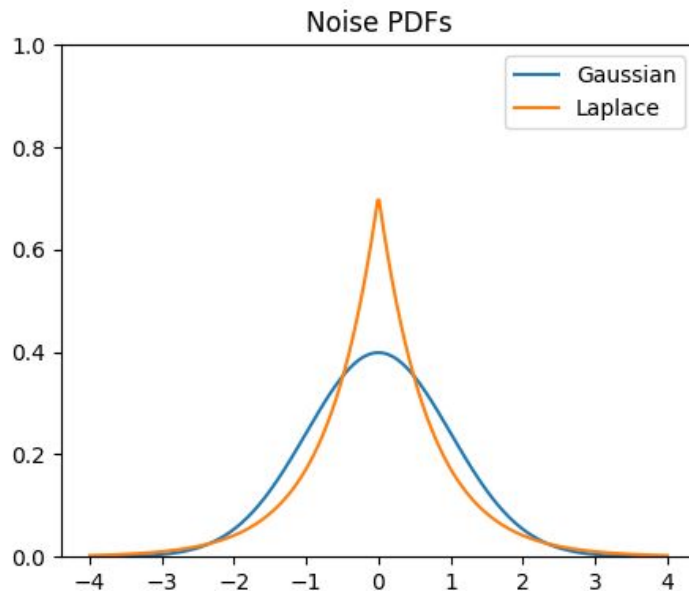
# Noise Model $\Leftrightarrow$ Error Function

## Zero-mean Gaussian Noise

$$p(\epsilon) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right)$$

$$-\log p(\epsilon) = \frac{\epsilon^2}{2\sigma^2} + \text{const}$$

$\Rightarrow$  Sum of the squares error

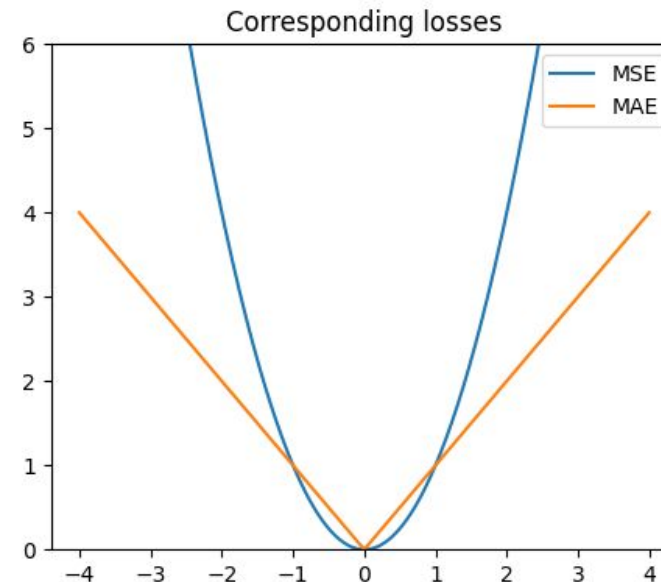


## Zero-mean Laplacian Noise

$$p(\epsilon) = \frac{1}{2b} \exp\left(-\frac{|\epsilon|}{b}\right)$$

$$-\log p(\epsilon) = \frac{|\epsilon|}{b} + \text{const}$$

$\Rightarrow$  Sum of the Absolute error





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# Prior Beliefs

- MLE Recap: Least Squares as Maximum Likelihood
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# Recall: Beliefs and Priors

In the coin flip example, assume the true probability of head is  $\theta$ .

- **Frequentist:** Use data only (MLE).

$$\hat{\theta} = \frac{\#heads}{\#flips}$$

Flip twice, see HH, then:

$$\hat{\theta}_{MLE} = \frac{2}{2} = 1$$

Overconfident with  $n = 2!!!$

- **Bayesian:** Combine data with belief.

- Prior belief:  $h_0 = t_0 = 5$

- See HH:

- Posterior  $p(\theta|HH) \propto p(\theta) \times \text{likelihood}$

Strong prior ensures  
alignment with beliefs

Chases the data  
 $p(HH|\theta) = \theta^2$

$$\hat{\theta}_{posterior\ mean} = \frac{5 + 2}{10 + 2} = 0.58$$

0 ——— 0.5 (prior) ——— 0.58 (Posterior mean) ——— 1.0 (MLE)



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# What About Regression? MLE for Weights: No Prior

- In regression, MLE finds weights  $w$  by maximizing likelihood of data under noise model.
- **Problem:** with few/noisy data points  $\rightarrow$  can fit extreme weights.
- No mechanism to encode “weights should be small/reasonable.”
- Why small?
  - With inputs in SD units we don't expect huge effects unless data is persuasive.
  - Shrinking  $w$  lowers prediction variance, especially when features are correlated.



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# Belief about Parameters

- Just like coins: we can encode a prior belief about weights  $w$ .
- Common prior:  $w_j \sim \mathcal{N}(0, \sigma_w^2)$ 
  - Centered at 0  $\rightarrow$  no strong effect unless data supports it.
  - Variance  $\sigma_w^2$  controls how strongly we expect small weights.
- This prior = belief that simpler models are preferred.

Likelihood function

$$p(\mathbf{t} | \mathbb{X}, w, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | w^T \phi(\mathbf{x}_n), \sigma^2)$$

Prior (small, centered)

$$p(w | \tau^2) = \frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|w\|^2}{2\tau^2}}$$

Posterior (from Bayes rule) up to a constant:

$$p(w | \mathbf{t}) \propto$$



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# What Is MAP?

Bayes Rule

$$p(\mathbf{w}/\mathbf{t}) \propto \underbrace{\prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)}_{\text{Likelihood}} \underbrace{\frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|\mathbf{w}\|^2}{2\tau^2}}}_{\text{Prior}}$$

Maximum A Posteriori (MAP):

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w}/\mathbf{t}) = \underset{\mathbf{w}}{\operatorname{argmax}} [p(\mathbf{t}|\mathbf{w}) \times p(\mathbf{w})]$$



# What Is MAP?

Maximum A Posteriori (MAP):

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmax}} p(w/t) = \underset{w}{\operatorname{argmax}} [p(t|w) \times p(w)]$$

Taking  $-\log$  of the objective turns maximization problem to a minimization problem:

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} [-\ln p(t|w) - \ln p(w)]$$



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# Plugging in Distributions

$$\begin{aligned}\hat{w}_{MAP} &= \underset{w}{\operatorname{argmin}} \left[ \underbrace{-\ln p(\mathbf{t} | w)}_{-\ln \prod_{n=1}^N \mathcal{N}(t_n | w^T \phi(x_n), \sigma^2)} - \underbrace{\ln p(w)}_{-\ln \frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|w\|^2}{2\tau^2}}} \right] \\ &= \boxed{\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(x), w))^2} = \boxed{\frac{\|w\|^2}{2\tau^2}} + \text{const} \\ \hat{w}_{MAP} &= \underset{w}{\operatorname{argmin}} \left[ \boxed{\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(x), w))^2} + \boxed{\frac{\|w\|^2}{2\tau^2}} \right]\end{aligned}$$



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# Does This Look Like a Ridge Regression?

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \left[ \frac{1}{2\sigma^2} \sum_n (\mathbf{t}_n - y(\phi(\mathbf{x}), \mathbf{w}))^2 + \frac{\|\mathbf{w}\|^2}{2\tau^2} \right]$$

Multiply by  $\sigma^2$

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \left[ \underbrace{\frac{1}{2} \sum_n (\mathbf{t}_n - y(\phi(\mathbf{x}), \mathbf{w}))^2}_{\frac{1}{2} E_D(\mathbf{w})} + \underbrace{\frac{\sigma^2 \|\mathbf{w}\|^2}{2\tau^2}}_{E_w(\mathbf{w})} \right] \rightarrow \lambda$$

Least-Squares + Gaussian Prior  $\Rightarrow$  Ridge Regression



# What is the relationship between MAP and ridge regression?

# Least-Squares + Gaussian Prior $\Rightarrow$ Ridge Regression



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Posterior (Bayes Rule):  $p(\mathbf{w}|\mathbf{t}) \propto p(\mathbf{t}|\mathbf{w}) p(\mathbf{w})$

Work with the  $-\log$  :  $-\log(p(\mathbf{w}|\mathbf{t})) = -\log(p(\mathbf{t}|\mathbf{w})) - \log(p(\mathbf{w})) + \text{const}$

Plugging in Gaussian:

$$p(\mathbf{w}|\tau^2) = \frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|\mathbf{w}\|^2}{2\tau^2}} \quad \longrightarrow \quad \log(p(\mathbf{w})) = \frac{\|\mathbf{w}\|^2}{2\tau^2} + \text{const}$$
$$-\log(p(\mathbf{t}|\mathbf{w})) = \frac{1}{2\sigma^2} \sum_n (\mathbf{t}_n - y(\phi(\mathbf{x}), \mathbf{w}))^2$$

Combine and multiply by  $2\sigma^2$

$$\mathbf{w}_{MAP}^* = \arg \min_{\mathbf{w}} \sum_n (\mathbf{t}_n - y(\phi(\mathbf{x}), \mathbf{w}))^2 + \frac{\sigma^2 \|\mathbf{w}\|^2}{\tau^2}$$

$\lambda$  for ridge regression

# Bias-Variance Trade-off

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# Fundamental Challenges in Learning?



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- ***Fit the Data***
  - Provide an explanation for what we observe
- ***Generalize to the World***
  - Predict the future
  - Explain the unobserved



Is this cat grumpy or are we  
overfitting to human faces?



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# Fundamental Challenges in Learning?

- **Bias:** The expected deviation between the predicted value and the true value.
- **Variance:** Two sources
  - **Noise:** The variability of the random noise in the process we are trying to model.
  - **Model Variance:** The variability in the predicted value across different training datasets.

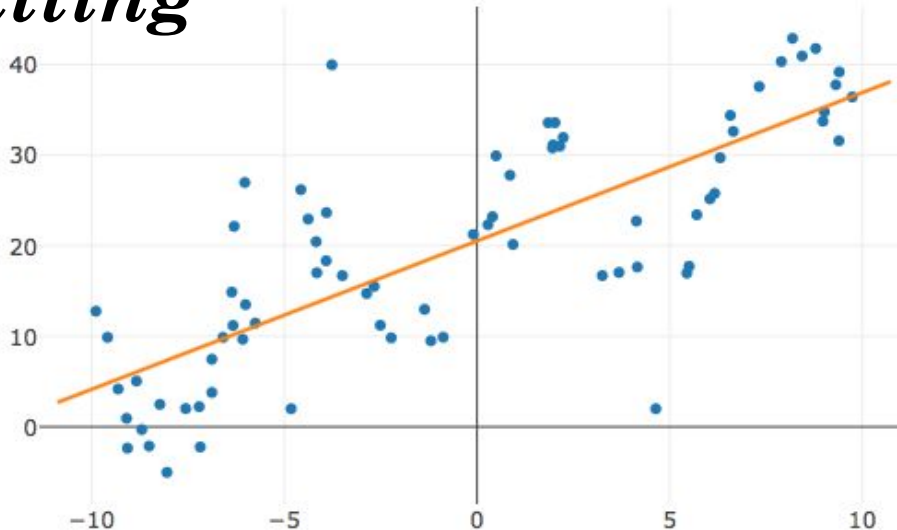
# Bias



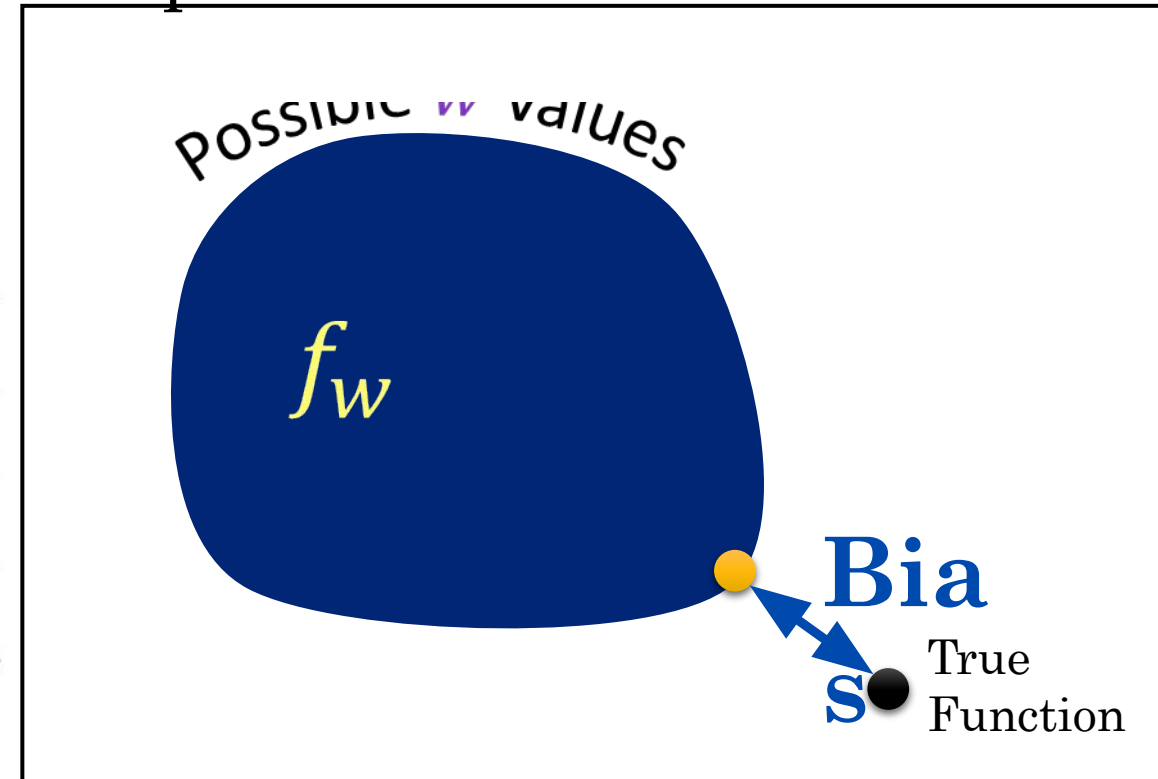
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*The expected deviation between the predicted value and the true value*

- Depends on both the:
  - Choice of  $f$
  - Learning procedure
- ***Under-fitting***



All possible functions



# Noise

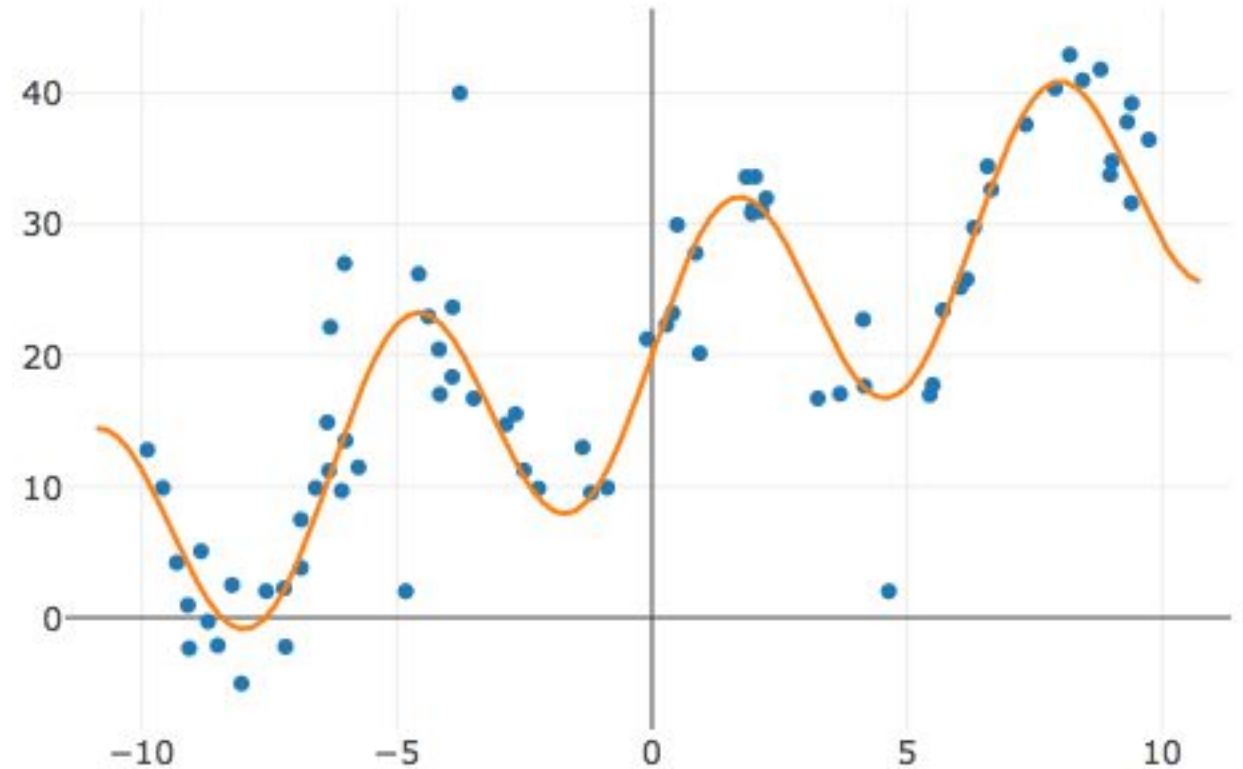


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The variability of the random noise in the process we are trying to model.

- Measurement variability
- Stochasticity
- Missing information

**Beyond our control  
(usually)**



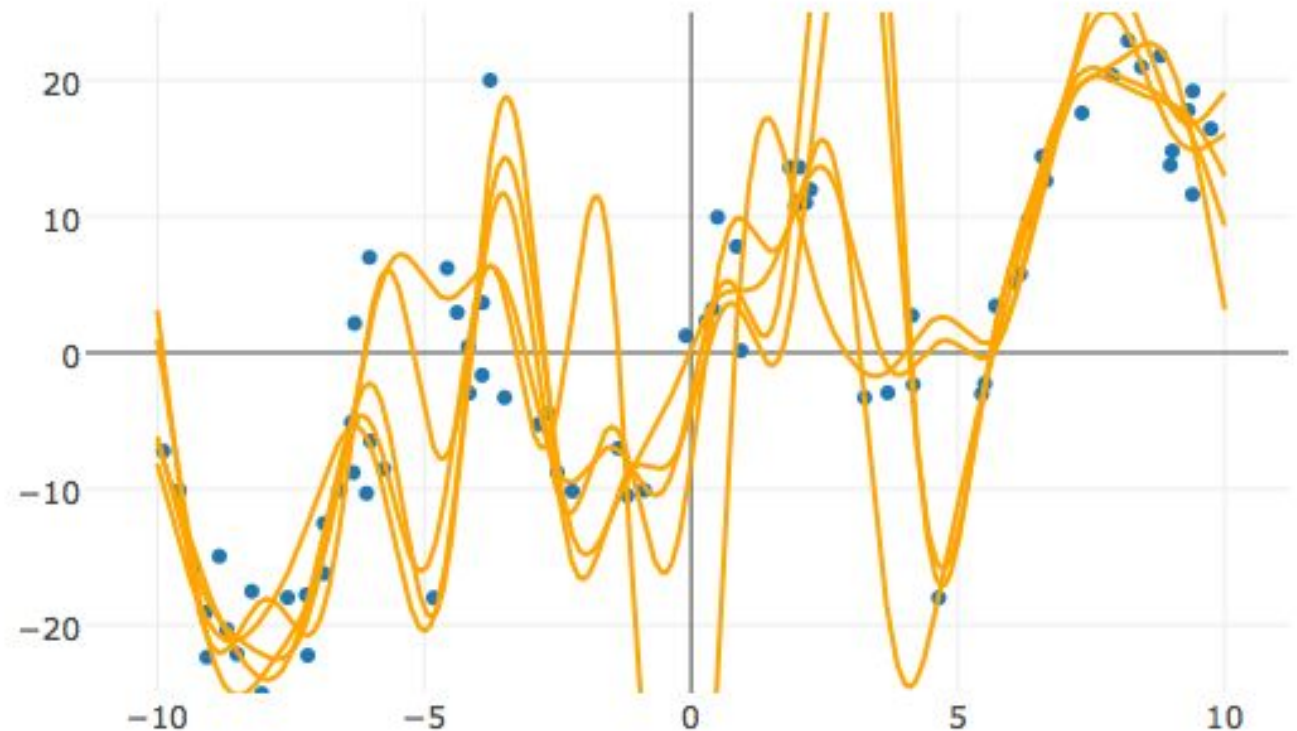


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# Model Variance

*Variability in the predicted value across different training datasets.*

- Sensitivity to variation in the training data
- Poor generalization
- **Overfitting**





**Which of the following models  
would have high bias?**



**Which of the following models  
would have high variance?**

# Analysis of Squared Error



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- For the test point  $\mathbf{x}$  the expected error:
  - *Random variables are red*

Assume noisy observations  
→  $t$  is a random variable

True Function

$$t = h(\mathbf{x}) + \epsilon$$

Noise term:

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

Can be any  
parametric function

Assume **training data** is random  
→  $w$  is a random variable

# Analysis of Squared Error



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Goal:

$$E[(t - f_w(x))^2] =$$

**“Noise” + (Bias)<sup>2</sup> + Model Variance**



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$$E[(t - f_w(\mathbf{x}))^2] = E[(t - \underbrace{h(\mathbf{x}) + h(\mathbf{x})}_{\text{Subtracting and adding } h(\mathbf{x})} - f_w(\mathbf{x}))^2]$$

Subtracting and adding  $h(\mathbf{x})$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2] = E[\underbrace{(t - h(\mathbf{x}))}_a + \underbrace{h(\mathbf{x}) - f_w(\mathbf{x}))}_b]^2]$$

Expanding in terms of  $a$  and  $b$ :

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$\begin{aligned} &= E\left[\underbrace{(t - h(\mathbf{x}))^2}_{a^2}\right] + E\left[\underbrace{(h(\mathbf{x}) - f_w(\mathbf{x}))^2}_{b^2}\right] \\ &\quad + 2 E\left[\underbrace{(t - h(\mathbf{x}))}_{\epsilon} \underbrace{(h(\mathbf{x}) - f_w(\mathbf{x}))}_{2ab}\right] \\ &\quad + 2 E[\underbrace{\epsilon}_{\epsilon} (h(\mathbf{x}) - f_w(\mathbf{x}))] \end{aligned}$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$E[(t - f_w(x))^2] = E[\underbrace{(t - h(x))}_a + \underbrace{h(x) - f_w(x)}_b]^2]$$

Expanding in terms of  $a$  and  $b$ :

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$= E[a^2] + E[b^2] + 2 E[a b]$$

Independence of  $\epsilon$  and  $w$

$$+ 2 \underbrace{E[\epsilon]}_0 E[h(x) - f_w(x)]$$

Useful Equations:

$$t = h(x) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2]$$

 $\hat{=}$ 

$$= E[(t - h(\mathbf{x}))^2] +$$

Obs. Value

True Value

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

True Value

Pred. Value

“*Noise*” Term

Model  
Estimation  
Error

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \overset{\sigma^2}{\parallel}$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model  
Estimation  
Error

We need to calculate  
this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model Estimation Error

We need to calculate  
this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



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$$\mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] = \text{Next we will show....}$$

$$\left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 + \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right]$$

**(Bias)  
2**

**Model  
Variance**

□ How?

□ Adding and subtracting what?



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$$\mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] =$$

$$\mathbb{E} \left[ \underbrace{\left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_a + \underbrace{\left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)}_b \right]^2$$

Expanding in terms of  $a$  and  $b$ :

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$\begin{aligned} & \mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] + \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ & 2\mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right] \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[ (h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}))^2 \right] &= \\ \mathbb{E}[(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})])^2] &+ \mathbb{E} \left[ (\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}))^2 \right] + \\ 2\mathbb{E}[(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})]) &(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}))] \\ \underbrace{\hspace{10em}}_{\text{constant}} & \\ \underbrace{2(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})])}_{\text{constant}} &\underbrace{\mathbb{E}[(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}))]}_{\text{constant}} \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] &+ \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ 2 \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) &\underbrace{\mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right]}_{\text{constant}} \\ \underbrace{2 \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_{\text{constant}} &\underbrace{\left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_0 \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \mathbb{E} \left[ \underbrace{\left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2}_{\text{constant}} \right] &+ \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] \\ &= \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 + \end{aligned}$$



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$$\mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] = \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 + \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right]$$

**(Bias)  
2**

**Model  
Variance**

$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \overset{\sigma^2}{\parallel}$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model  
Estimation  
Error

We now have  
calculated this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

$\approx \sigma^2$

$$= E[(t - h(\mathbf{x}))^2] + \text{“Noise” Term}$$

$$(h(\mathbf{x}) - E[f_w(\mathbf{x})])^2 + \text{(Bias)}$$

$\frac{1}{2}$

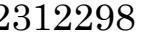
$$E[(E[f_w(\mathbf{x})] - f_w(\mathbf{x}))^2] \text{ Model Variance}$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$

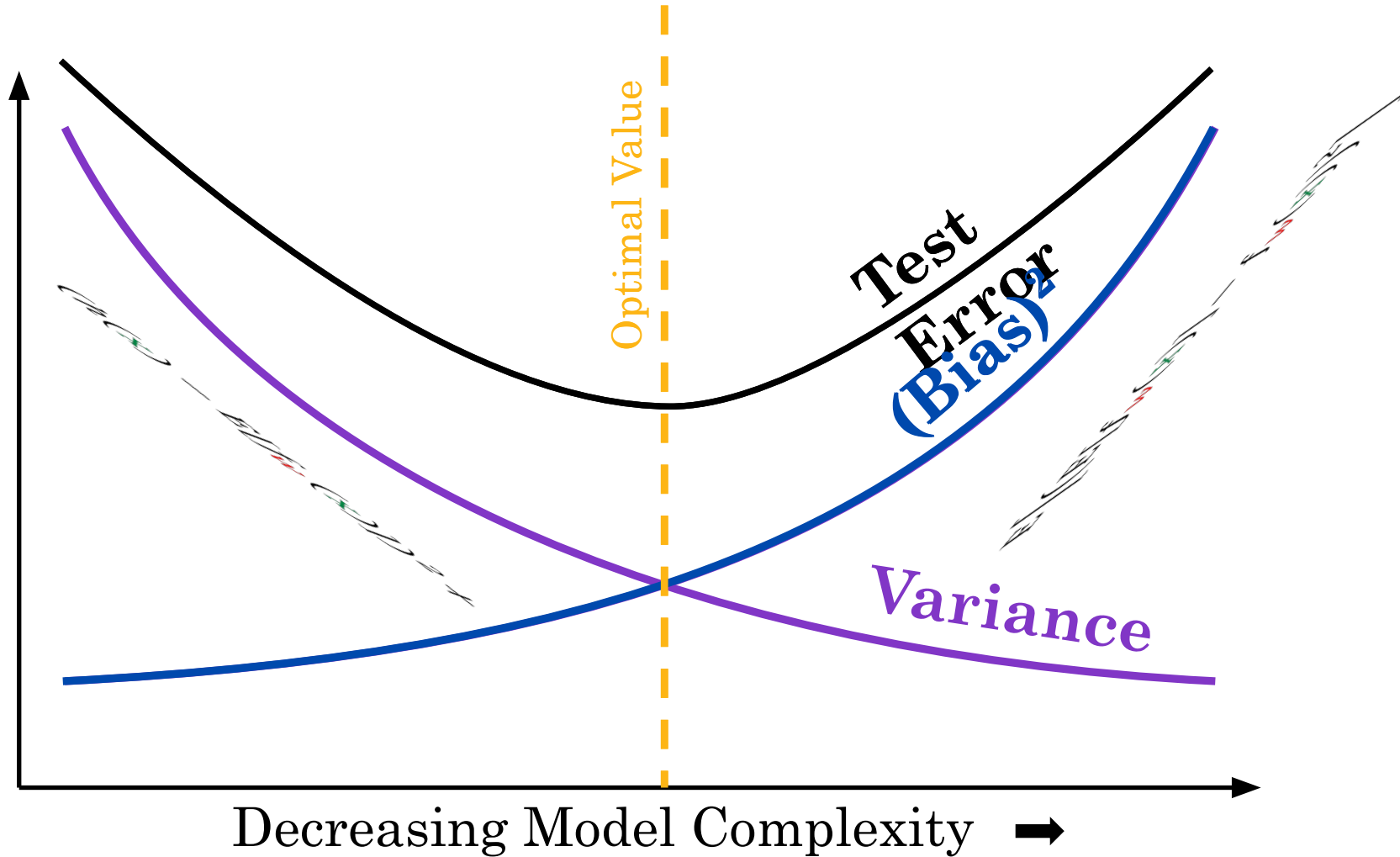




# Bias Variance Plot



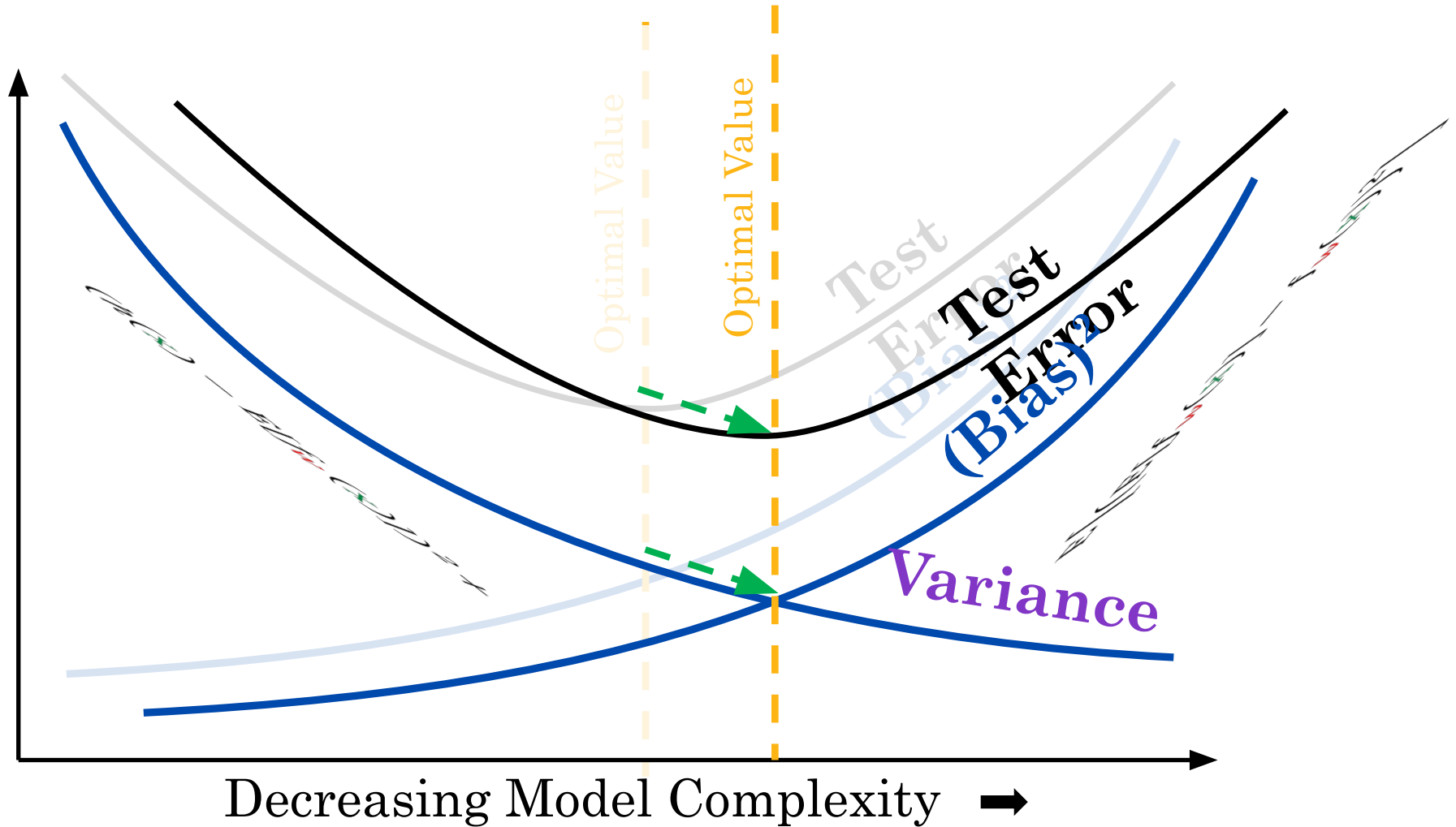
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# More Data Supports More Complexity



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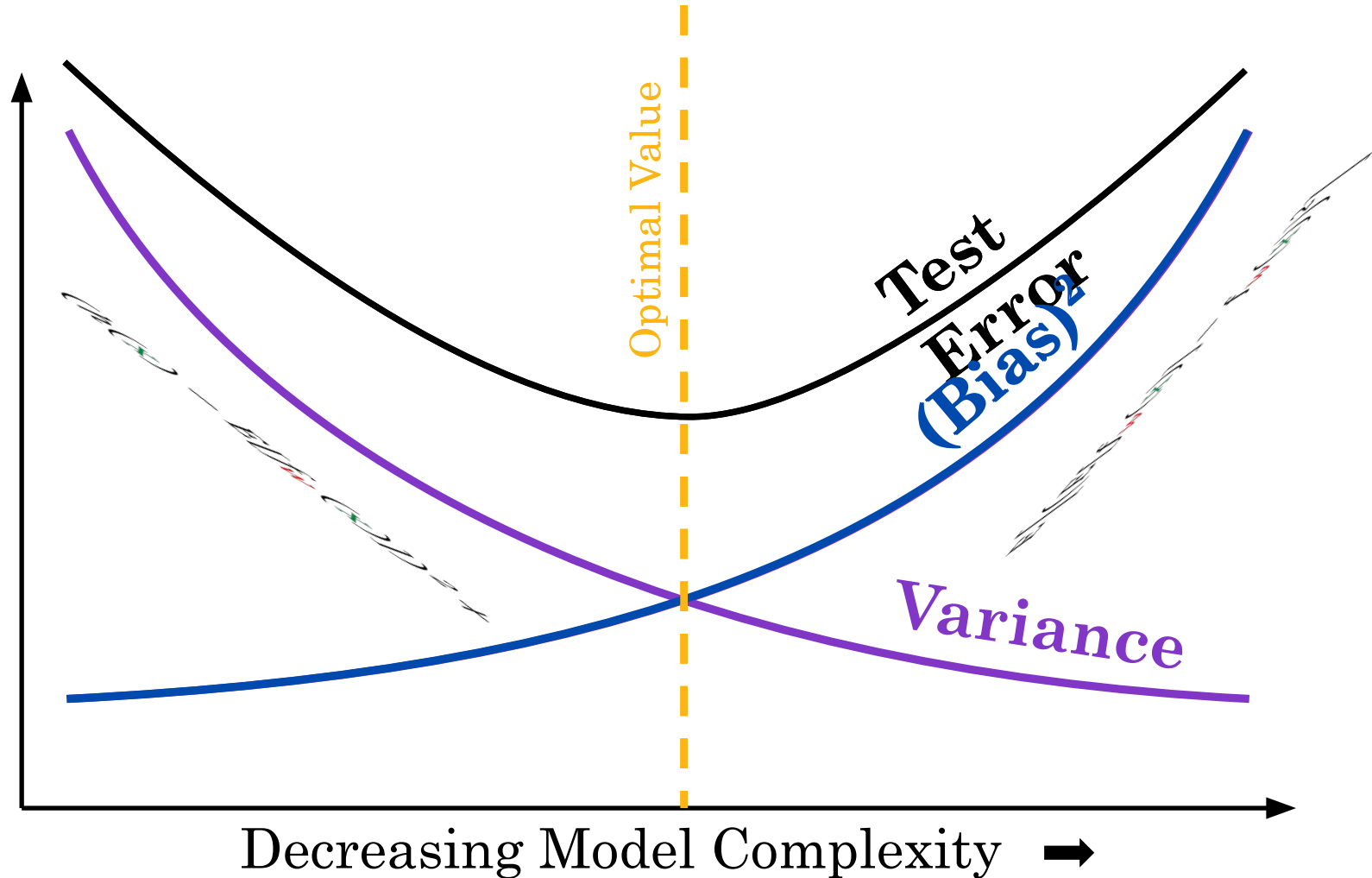


# How Do We Control Model Complexity?

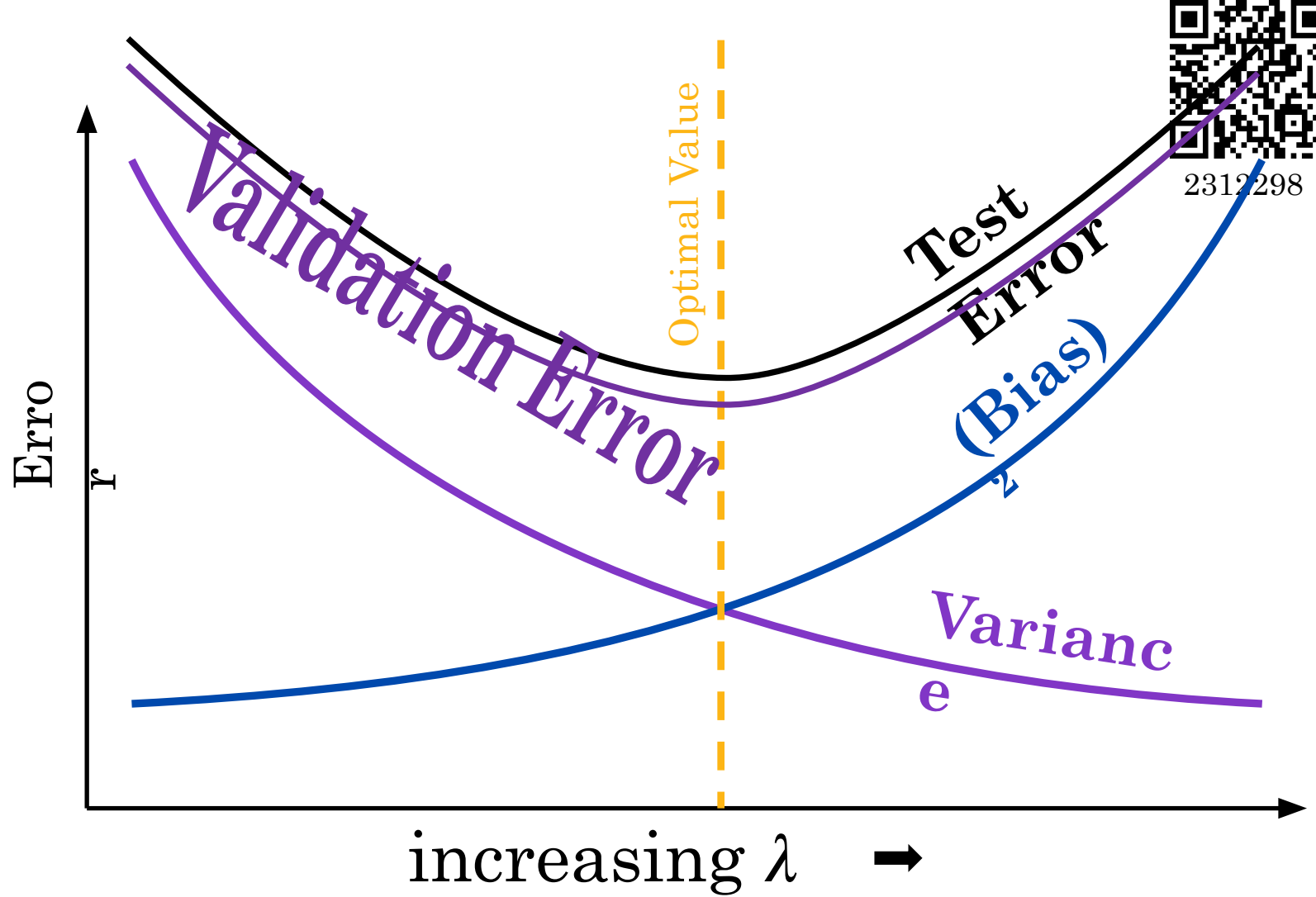


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- So far:
  - Number of features
  - Choices of features
  - **Regularization**



# Determining the Optimal $\lambda$



- Value of  $\lambda$  determines bias-variance tradeoff
  - Larger values  $\square$  more regularization  $\square$  more bias  $\square$  less variance
  - $\square$  Determined through validation



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# Dataset Example

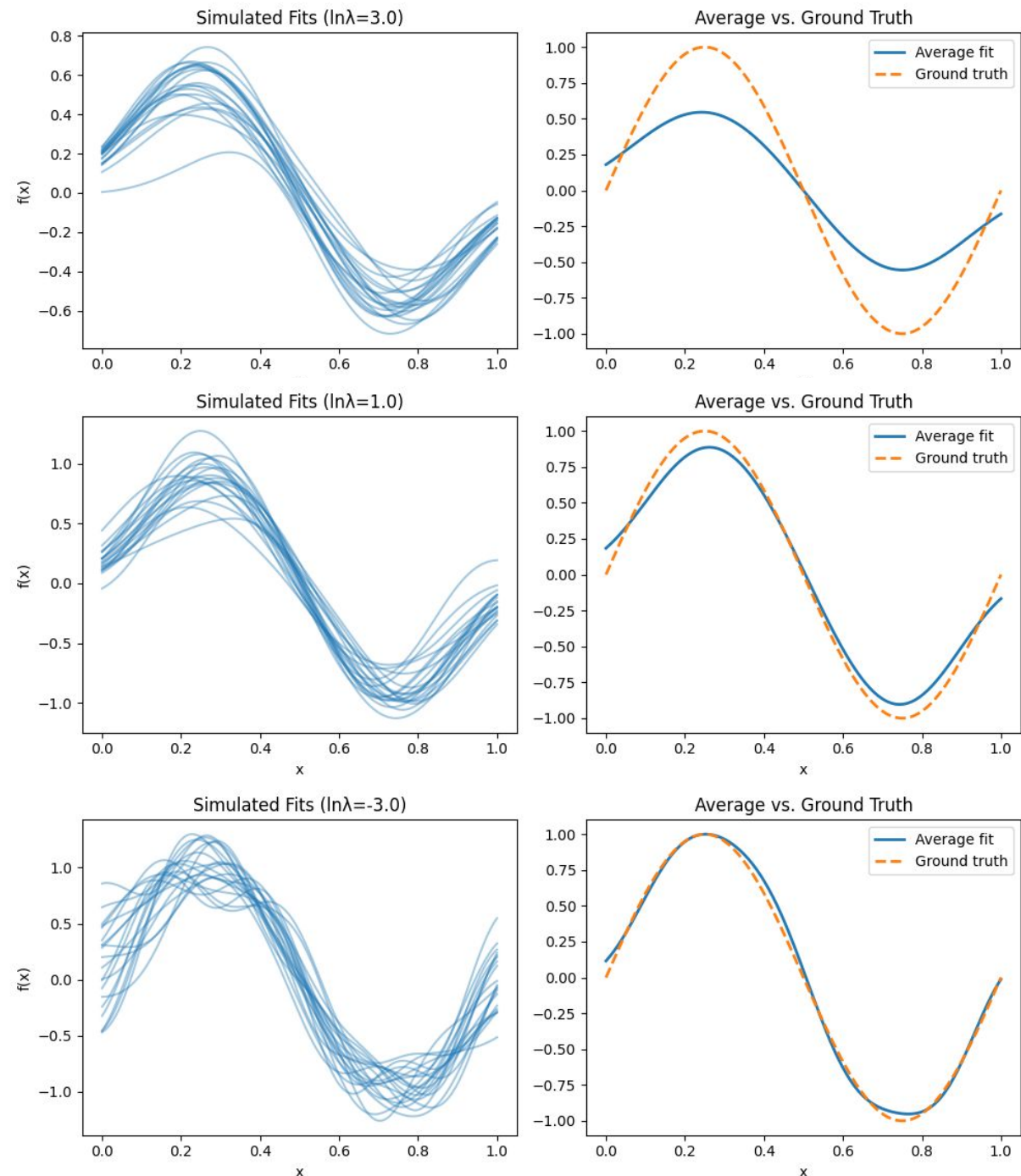
**Data generation function:**

$$h(x) = \sin(2\pi x), x \sim \text{Uniform}(0,1)$$

**Generate**  $L$  independent datasets ( $L = 100$ ), each with  $N = 25$  noisy observations.

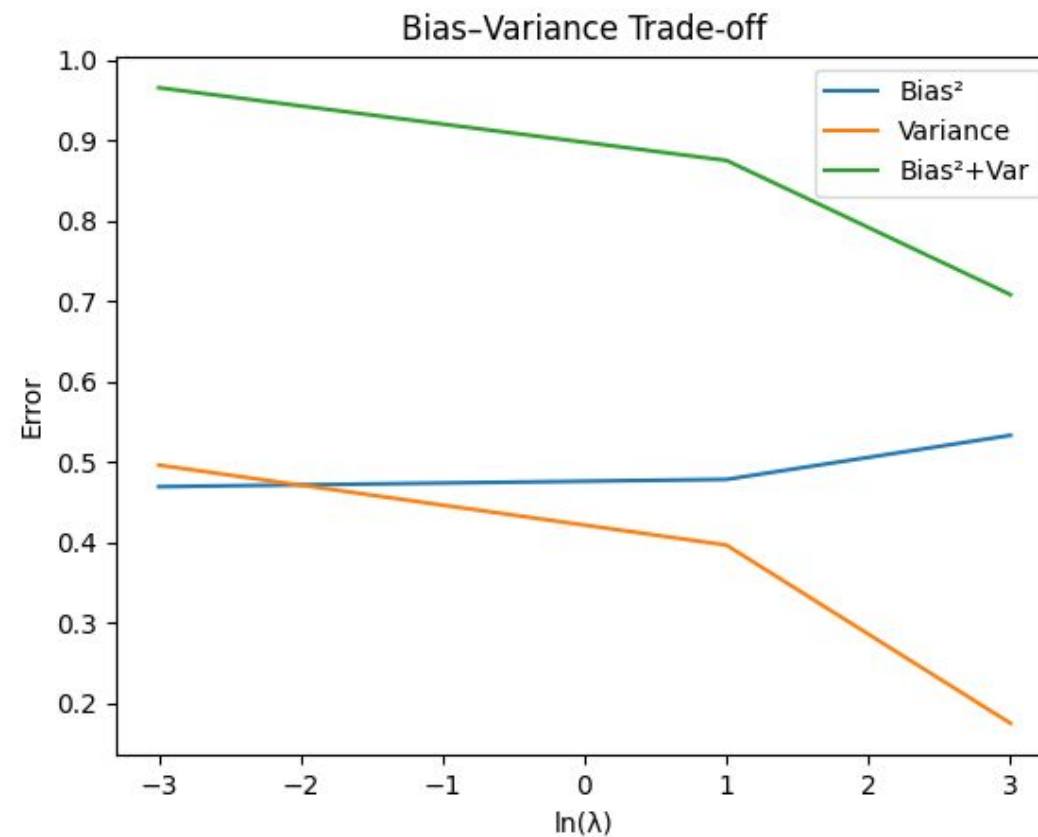
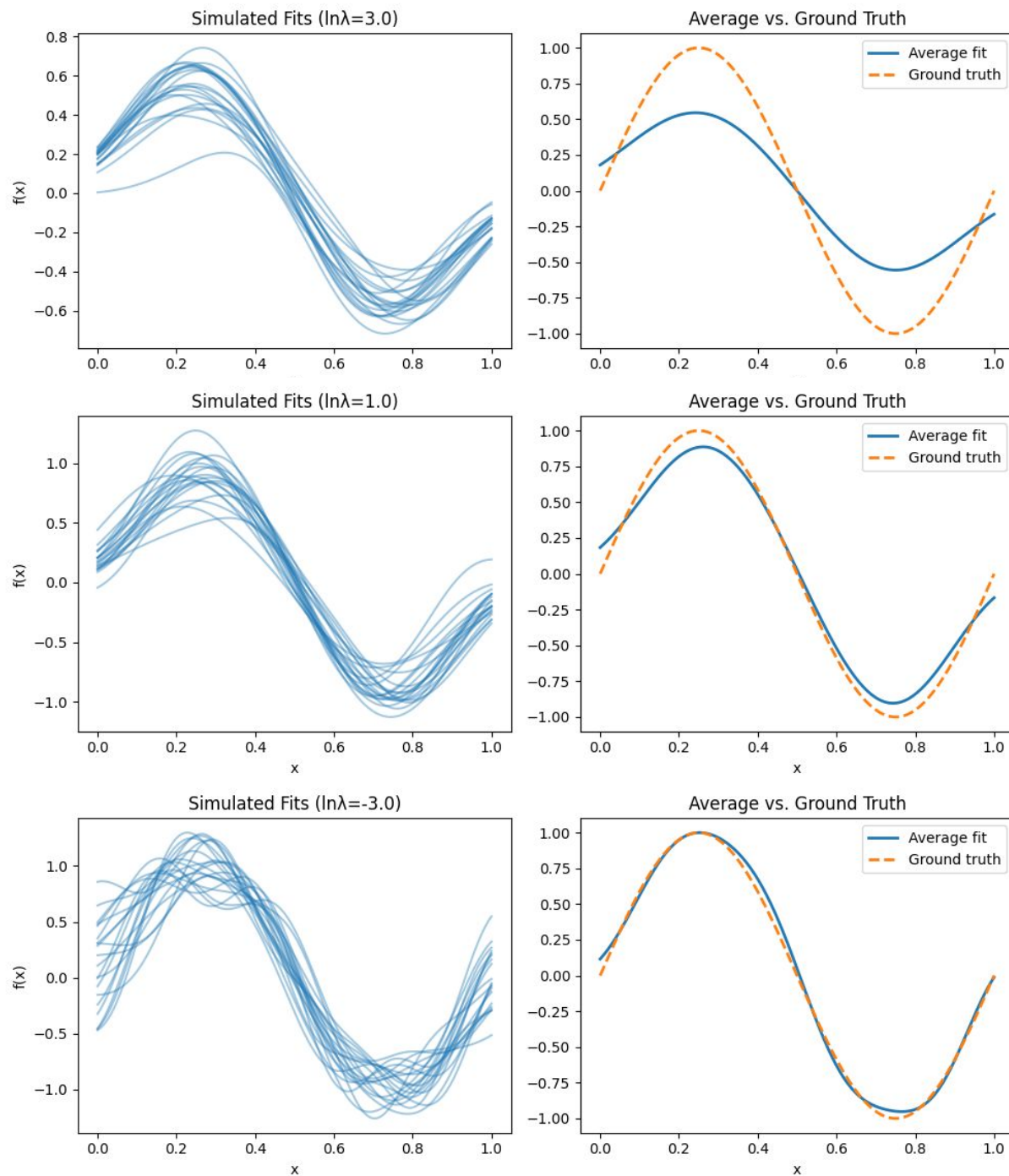
$$t_n = h(x_n) + \epsilon_n, \\ \epsilon_n \sim \mathcal{N}(0, 0.32)$$

**Model:** linear combination of  $M = 24$  Gaussian basis functions plus bias.





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**Select options below.**

# Bias Variance Derivation Quiz



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What is the equivalent of  $E[t]$ ?

- A. 0
- B. Bias
- C. Variance
- D. Noise
- E.  $h(\mathbf{x})$
- F.  $h(\mathbf{x}) + \epsilon$



**Select options below.**

# Bias Variance Derivation Quiz



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What is the equivalent of  $E[\epsilon^2]$ ?

- A. 0
- B. Bias
- C. Variance
- D. Noise
- E.  $h(\mathbf{x})$
- F.  $h(\mathbf{x}) + \epsilon$



**Select options below.**



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# Bias Variance Derivation Quiz

What is the equivalent of  $E \left[ \left( E[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right]$ ?

- A. 0
- B. Bias
- C. Variance
- D. Noise
- E.  $h(\mathbf{x})$
- F.  $h(\mathbf{x}) + \epsilon$

## Lecture 9

# Regression (Cont.) and Bias-Variance Trade-off

**Credit:** Joseph E. Gonzalez and Narges Norouzi

**Reference Book Chapters:** Chapter 4.2 and 4.3



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# Homework!

# HW 1 Updates



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- **Please check ed for any HW guidance.** We have added a walkthrough for submitting your homework to gradescope
- Autograder for problem 9 of part 2 has been adjusted to give partial credit (you should receive updated grades this afternoon):
  - 38-40%: 1 pt
  - 40-42: 2 pts
  - 43-44: 3 pts
  - >44: 5 pts
- There is a separate assignment on Gradescope for just your PDF export – this will be the norm moving forward
- **Thank yall for being so patient!**

# HW2



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## Part 1 (Due Oct 3rd)

- Written problems for regression and MLE
- Short coding portion to explore LMArena (formerly Chatbot Arena) data

Uses concepts from  
Logistic regression  
lectures next week



## Part 2 (Due Oct 17th)

- Paper questions (tutorial on how to read a paper next week)
  - Chatbot Arena
  - VibeCheck / Style control
- Coding implementation of Chatbot Arena evaluation and simplified version of VibeCheck

# Data Visualizers for HW2

**Gradio** – a python backed UI developer created by huggingface

In the hw we encourage you to play around with the data you visualize and the styles of the UI

**YOU SHOULD VIBE CODE THESE SINCE IT IS EASY TO VERIFY**

Spaces BerkeleyML/Chatbot-Arena-Viewer like 0 Building

App Files Community Settings

### Chatbot Arena Battle Viewer (100k)

Filter by **Model A**, **Model B**, and **Language**, then browse side-by-side conversations. Data: lmarena-ai/arena-human-preference-100k.

**Model A** gemma-2-27b-it **Model B** gemini-1.5-pro-api-0514 **Language** English

Random match ◀ Prev Next ▶

**Model A: gemma-2-27b-it**

User

Mr Stone was buying firehouse logs. The bill was ninety dollars. He gave the salesman what he thought were two \$50 notes, (actually two \$100 notes). The salesman was sleepy and didn't notice either, so he gave Mr Stone what he thought was a \$10 note (actually three \$50 notes). Who ended up better off than they should?

Assistant

Mr. Stone ended up better off than he should have. Here's why:

- **What Mr. Stone should have paid:** \$90
- **What Mr. Stone actually paid:** \$200 (two \$100 bills)
- **What the salesman gave Mr. Stone back:** \$150 (three \$50 bills)

**Mr. Stone effectively received \$60 (\$150 - \$90) more than he should have.**

**Model B: gemini-1.5-pro-api-0514**

User

Mr Stone was buying firehouse logs. The bill was ninety dollars. He gave the salesman what he thought were two \$50 notes, (actually two \$100 notes). The salesman was sleepy and didn't notice either, so he gave Mr Stone what he thought was a \$10 note (actually three \$50 notes). Who ended up better off than they should?

Assistant

Mr. Stone ended up **much** better off than he should have. Here's why:

- **Mr. Stone paid:** \$200 (two \$100 notes)
- **Mr. Stone should have paid:** \$90
- **Mr. Stone received in change:** \$150 (three \$50 notes)
- **Mr. Stone should have received in change:** \$10

**Let's break it down:**

Mr. Stone paid \$200 for the logs, but he effectively paid \$90 for the logs, so he effectively received \$110 for the logs.



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Find these viewers on our Huggingface Page →





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## Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference

Wei-Lin Chiang<sup>\*1</sup> Lianmin Zheng<sup>\*1</sup> Ying Sheng<sup>2</sup> Anastasios N. Angelopoulos<sup>1</sup> Tianle Li<sup>1</sup> Dacheng Li<sup>1</sup>  
Banghua Zhu<sup>1</sup> Hao Zhang<sup>3</sup> Michael I. Jordan<sup>1</sup> Joseph E. Gonzalez<sup>1</sup> Ion Stoica<sup>1</sup>

### Abstract

Large Language Models (LLMs) have unlocked new capabilities and applications; however, evaluating the alignment with human preferences still poses significant challenges. To address this issue, we introduce Chatbot Arena, an open platform for evaluating LLMs based on human preferences. Our methodology employs a pairwise comparison approach and leverages input from a diverse user base through crowdsourcing. The platform has been operational for several months, amassing over 240K votes. This paper describes the platform, analyzes the data we have collected so far, and explains the tried-and-true statistical methods we are using for efficient and accurate evaluation and ranking of models. We confirm that the crowdsourced questions are sufficiently diverse and discriminating and that the crowdsourced human votes are in good agreement with those of expert raters. These analyses collectively establish a robust foundation for the credibility of Chatbot Arena. Because of its unique value and openness, Chatbot Arena has emerged as one of the most referenced LLM leaderboards, widely cited by leading LLM developers and companies. Our demo is publicly available at <https://chat.lmsys.org>.

| Evaluation Metric | Question Source  |                         |                            |
|-------------------|------------------|-------------------------|----------------------------|
|                   | Static           |                         | Live                       |
|                   | Ground Truth     | MMLU, HellaSwag, GSM-8K | Codeforces Weekly Contests |
|                   | Human Preference | MT-Bench, AlpacaEval    | Chatbot Arena              |

**Figure 1.** Classification of LLM benchmarks: We categorize along two dimensions: whether the questions are from a static dataset or a live, fresh source, and whether the evaluation metric relies on ground truth or (approximated) human preferences. MMLU (Hendrycks et al., 2020), HellaSwag (Zellers et al., 2019), GSM-8K (Cobbe et al., 2021), MT-Bench (Zheng et al., 2023b), and AlpacaEval (Li et al., 2023) are common examples of static benchmarks. Chatbot Arena is the platform introduced in this paper.

preferences in real-world, open-ended tasks.

To assess the performance of LLMs, the research community has introduced a variety of benchmarks. These benchmarks can be categorized based on two factors: the source of questions (either static or live) and the evaluation metric (either ground truth or human preference). According to these factors, benchmarks can be classified into four categories, as shown in Figure 1. While a range of benchmarks is beneficial, the most prevalent current method for evaluating LLMs remains a static, ground-truth-based evaluation, partly because such evaluations are inexpensive and reproducible.

However, these static, ground-truth-based benchmarks exhibit several limitations. Firstly, the questions within these benchmarks are not open-ended, hindering the ability to

## VIBECheck: DISCOVER & QUANTIFY QUALITATIVE DIFFERENCES IN LARGE LANGUAGE MODELS

Lisa Dunlap  
UC Berkeley

Krishna Mandal  
UC Berkeley

Trevor Darrell  
UC Berkeley

Jacob Steinhardt  
UC Berkeley

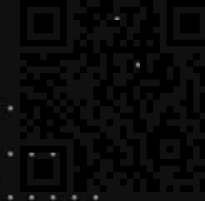
Joseph Gonzalez  
UC Berkeley

### ABSTRACT

Large language models (LLMs) often exhibit subtle yet distinctive characteristics in their outputs that users intuitively recognize, but struggle to quantify. These “vibes” – such as tone, formatting, or writing style – influence user preferences, yet traditional evaluations focus primarily on the singular vibe of correctness. We introduce **VibeCheck**, a system for automatically comparing a pair of LLMs by discovering identifying traits of a model (“vibes”) that are well-defined, differentiating, and user-aligned. VibeCheck iteratively discovers vibes from model outputs and then utilizes a panel of LLM judges to quantitatively measure the utility of each vibe. We validate that the vibes generated by VibeCheck align with those found in human discovery and run VibeCheck on pairwise preference data from real-world user conversations with Llama-3-70b vs GPT-4. VibeCheck reveals that Llama has a friendly, funny, and somewhat controversial vibe. These vibes predict model identity with 80% accuracy and human preference with 61% accuracy. Lastly, we run VibeCheck on a variety of models and tasks including summarization, math, and captioning to provide insight into differences in model behavior. VibeCheck discovers vibes like Command X prefers to add concrete intros and conclusions when summarizing in comparison to TNGL, Llama-405b often overexplains its thought process on math problems compared to GPT-4o, and GPT-4 prefers to focus on the mood and emotions of the scene when captioning compared to Gemini-1.5-Flash. Code and vibe visualizer found at <https://bench-mark.org/>

### 1 INTRODUCTION

**vibe check** : A process by which a group obtains a subjective assessment of another person, place, or thing.  
– *Urban Dictionary*



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# Evaluating LMM's in the Wild with



# LMArena

An open platform for  
human preference  
evals

# LMarena (Formerly Chatbot Arena)

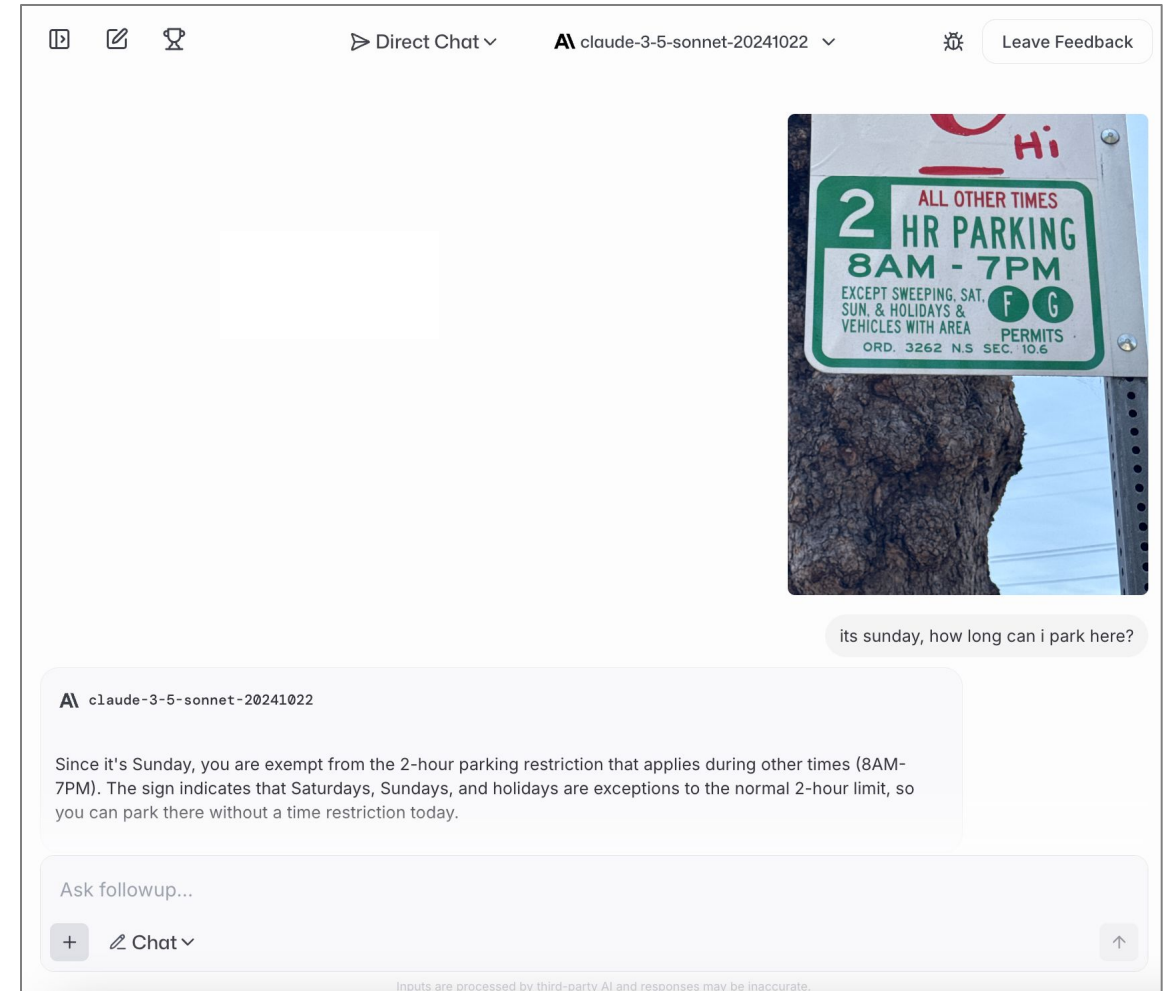
A platform for **holistic LMM evaluation**, where **real user conversations and pairwise votes** are crowdsourced to build a **live human preference leaderboard**.

**Members (pre-company launch):** Wei-Lin Chiang, Anastasios Angelopoulos, Lianmin Zheng, Ying Sheng, Lisa Dunlap, Chris Chou, Tianle Li, Evan Frick, Aryan Vichare, Naman Jain, Manish Shetty, Yifan Song, Kelly Tang, Sophie Xie, Connor Chen, Joseph Tennyson, Dacheng Li, Siyuan Zhuang, Valerie Chen, Wayne Chi  
**Advisors (pre-company launch):** Ion Stoica, Joseph Gonzalez, Hao Zhang, Trevor Darrell

# Crowdsourcing user interactions with LMM's

## Direct Chat

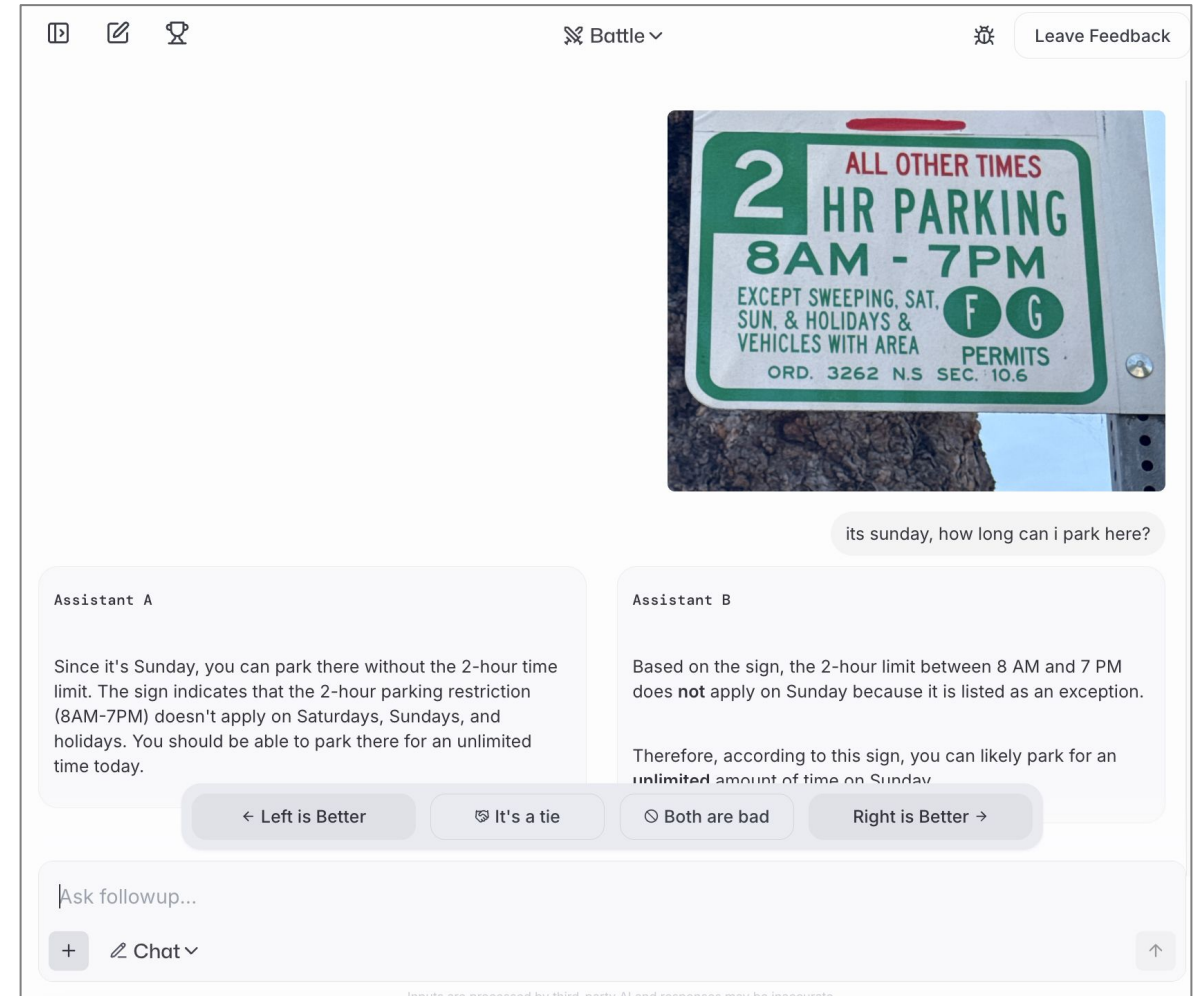
1. User selects a model to **chat** with (LLM, VLM, Text-2-Img, etc)
2. Conversations are recorded to **analyze** real world interactions

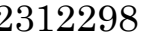


# Crowdsourcing user interactions with LMM's

## Battle

1. User input **any prompt**
  2. **Two** anonymized models give answers side-by-side
  3. User **vote** which is best
- Pairwise setting allows for more fine grained comparison



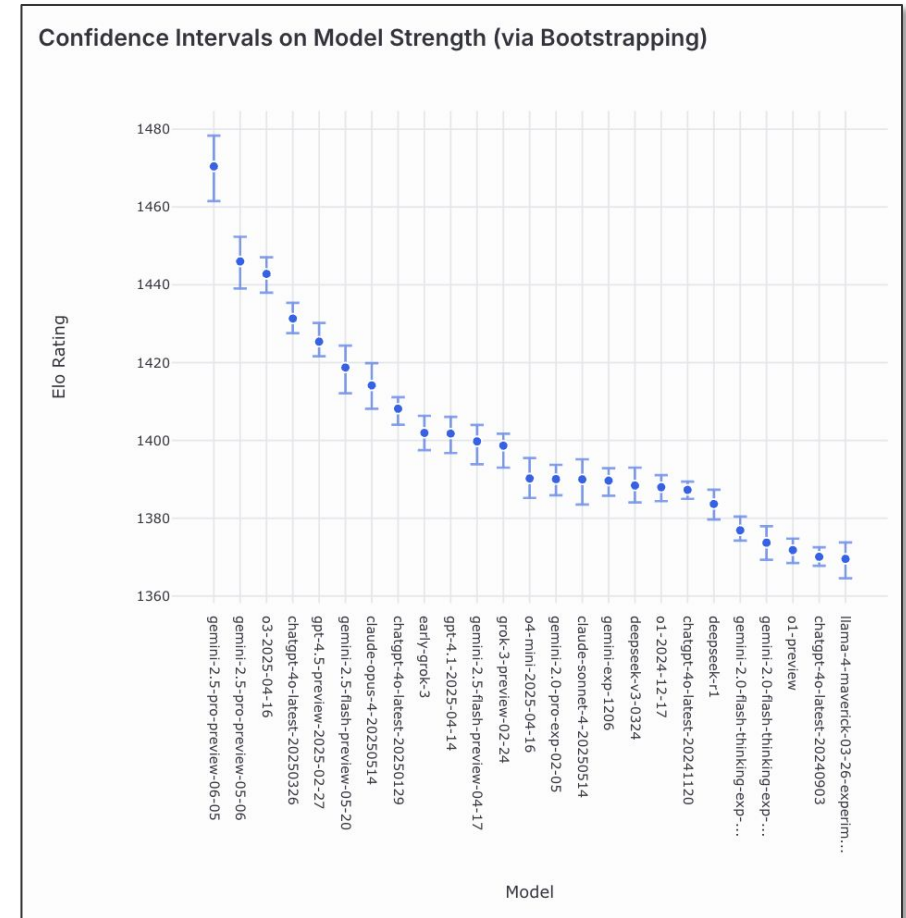


# Using Battles to Generate a Preference Leaderboard

- Bradley-Terry Model (next week):**  
estimate the relative strengths of  
models (**X**) through logistic regression  
(**Y** = winner)

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^M} \frac{1}{n} \sum_{i=1}^n \text{BCELoss}(\text{sigmoid}(X_i^\top \beta), Y_i)$$

The **Arena Score** of a model is it's coefficient in this model ( $\beta$ )



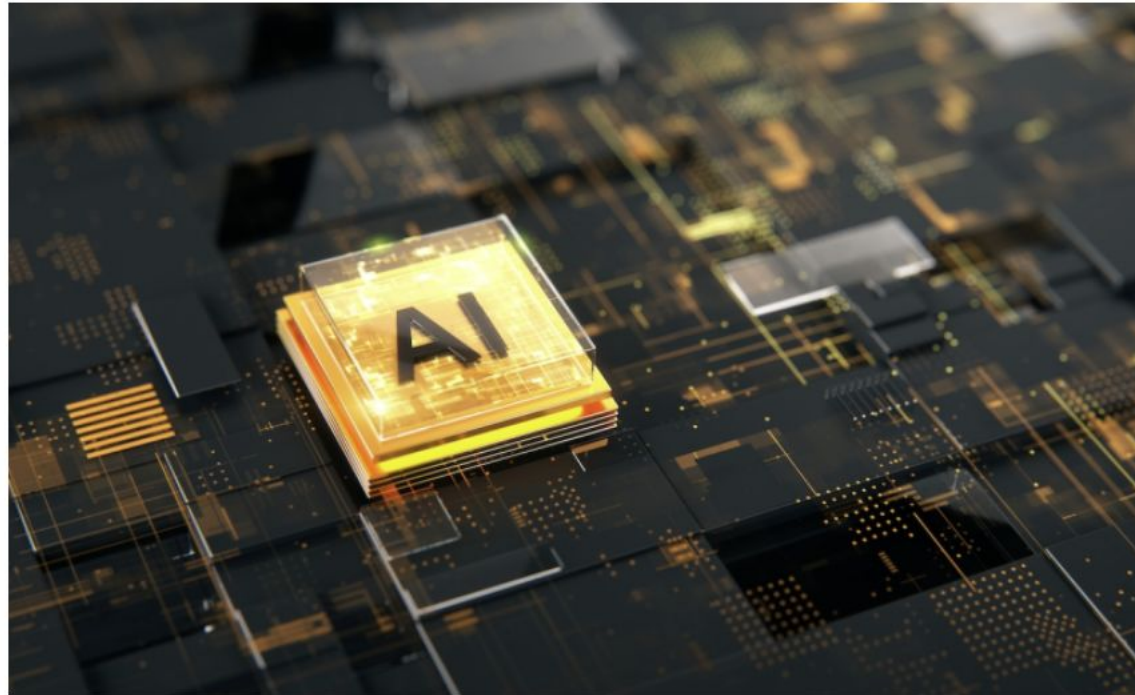
# If you think logistic regression is not valuable



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## Chatbot Arena Raises \$100 Million for Platform That Compares AI Models

BY PYMNTS | MAY 21, 2025



[Chatbot Arena](#) raised \$100 million in seed funding to support its platform that helps users compare the performance of different artificial intelligence models.

The company also said in a Wednesday (May 21) [post](#) on social platform X that it plans to relaunch the platform, [LMArena](#), with a new look based on feedback from members of its community.



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# Structure of a paper

- Abstract
- Intro
- Related works
- Methods and/or Problem Formulation
- Results
- Limitations and Conclusion



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# What to answer in every paper

1. What problem is this paper tackling?
2. What do prior works do and how do they fall short?
3. What is the key insight of this paper? What do they do that prior works do not?
4. What are the inputs and outputs of their method/contribution?
5. What are limitations to this method?

1-4 can usually be found in the intro



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# What to answer in every paper

1. What problem is this paper tackling? 1st paragraph
2. What do prior works do and how do they fall short? 1st - 2nd
3. What is the key insight of this paper? What do they do that prior works do not? 2nd- 4th
4. What are the inputs and outputs of their method/contribution? 2nd- 4th
5. What are limitations to this method?

1-4 can usually be found in the intro